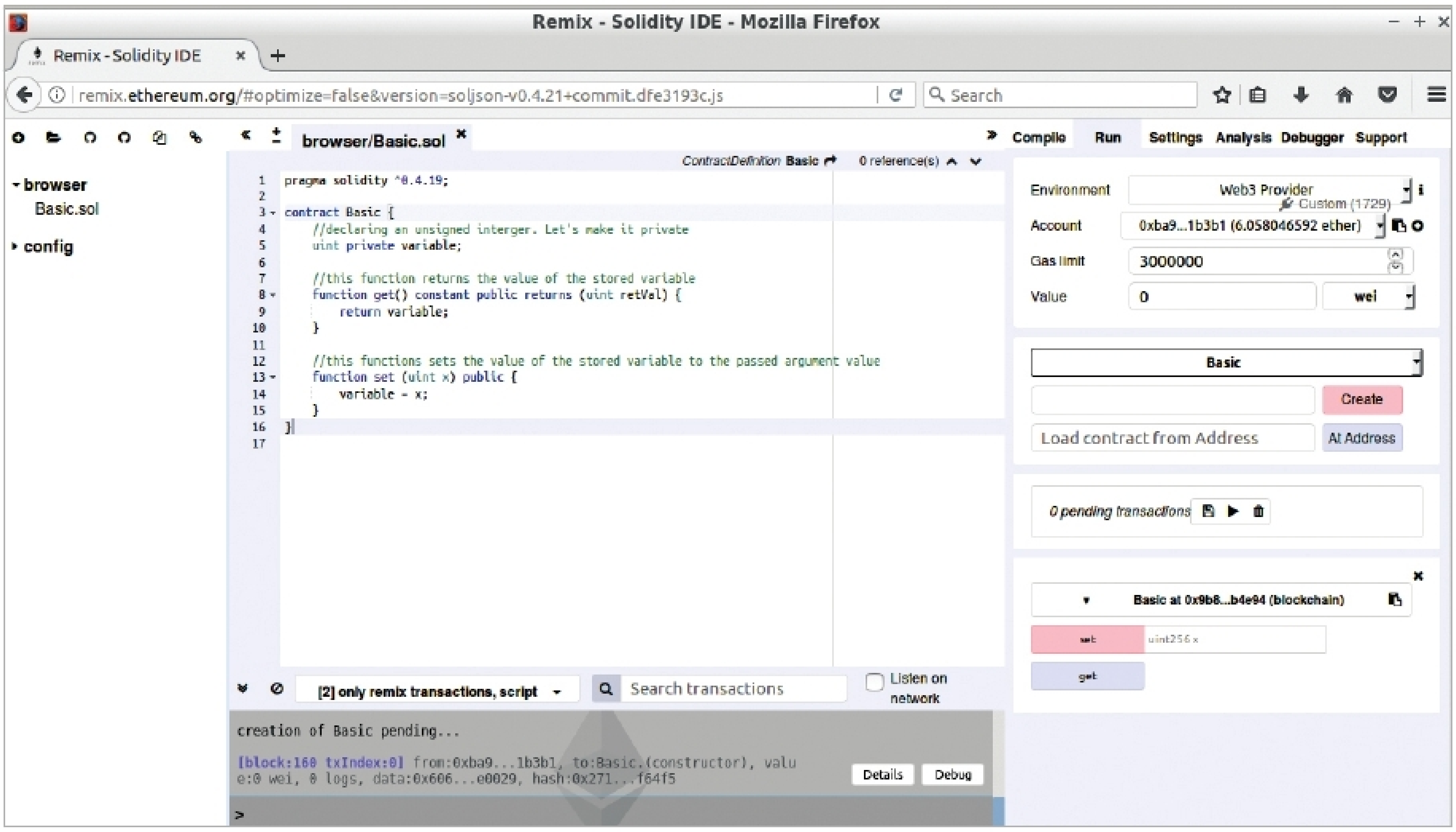

Figure 1: Using Remix to deploy a contract and interact

Use the 'get' button to fetch the current value of the variable. Use 'set' to put some value into the variable which you can check by using 'get' again. You will notice that 'get' gives the value instantaneously, whereas 'set' takes time. Also, on using 'set', a new transaction is seen on the Geth console on Terminal2. This is because 'get' is a query which just returns the current state recorded on the blockchain, whereas 'set' attempts to change the state. Any state changes must go through consensus, and hence we need to wait till the transaction is mined before we can see the changes across the nodes.

The final steps

This article was meant to initiate you into the world of Ethereum. There are a lot of things which you can try, and make complex blockchain based projects. I would encourage those interested to try out the following.

- *Exploring the Geth console and expanding the private network to multiple nodes:* You can try to make a node malicious or shut down abruptly, and see if and how it impacts the transactions.
- *Writing more elaborate smart contracts:* You can send and receive funds to and from smart contracts. Try some of the projects at *https://www.ethereum.org/*.
- Using the online compiler was one of the ways to deploy and interact. You can manually compile and deploy using ABI (*https://github.com/ethereum/go-ethereum/wiki/Contract-Tutorial*) or use a framework like Truffle (*http://truffleframework.com/*).
- Geth is one of the flavours implemented in Go. You can use other language implementations like C++ and Python, look into the code and contribute. They are all open source (*https://github.com/ethereum*). You can even use GUI clients like Mist (*https://github.com/ethereum/mist*).
- *Use the testnet and main Ethereum network to test and deploy contracts:* Warning: The main network involves real money. So be sure about what you are doing. Writing secure contracts is a tough job (*https://medium.com/new-alchemy/a-short-history-of-smart-contract-hacks-on-ethereum-1a30020b5fd*)!
- Take a look at other smart contract based projects and learn from them (*https://www.stateofthedapps.com/dapps/tab/most-viewed*).

Ethereum has a very active community. In case you face issues, a quick Web search should fetch most answers. If not, try their Gitter, Reddit and Stack Exchange platforms. I hope this article helped demystify some of the blockchain buzzwords and gave readers a feel of what blockchain is about. It is still a developing technology, and you can contribute both in terms of theory and implementation. END

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